

UEE Enterprise Specification v6.0 - Production Reference Implementation

UEE ENTERPRISE SPECIFICATION v6.0

1. PURPOSE

Production-ready reference implementation for the Universal Evolution Engine (UEE).

2. REFERENCE PLATFORM

Frontend:

- Next.js 15
- React 19
- TypeScript
- TailwindCSS
- Shadcn/UI

Backend:

- Node.js
- Python Services
- GraphQL Gateway

3. REPOSITORY STRUCTURE

/apps/web

/apps/api

/apps/agents

/packages/ui

/packages/ontology

/packages/workflows

/packages/knowledge-graph

/infrastructure

/docs

4. DOMAIN MODEL

Strategy

Objective

Capability

Value Stream

Initiative

Product

Feature

Application

API

Data Product

Technology

Risk

Control

Metric

Agent

5. POSTGRESQL DATA MODEL

Tables:

strategies

objectives

capabilities

products

applications

risks

metrics

users

agents

6. NEO4J KNOWLEDGE GRAPH

Nodes:

Strategy

Objective

Capability

Product

Application

API

DataProduct

Technology

Relationships:

REALIZED_BY

SUPPORTED_BY

USES

PRODUCES

ENABLES

7. AGENT RUNTIME

Anthropic Claude

LangGraph

Vector Memory

Knowledge Graph Memory

8. AGENT CONTRACTS

Input Schema

Context Schema

Reasoning Schema

Output Schema

9. LANGGRAPH WORKFLOWS

Executive Query

Strategy Analysis

Portfolio Optimization

Architecture Assessment

Value Realization

10. API CONTRACTS

REST APIs

GraphQL APIs

OpenAPI Specifications

11. EVENT SCHEMA

StrategyCreated

ObjectiveUpdated

CapabilityChanged

ProductReleased

RiskIdentified

MetricUpdated

12. SECURITY MODEL

Zero Trust

RBAC

ABAC

NIST Controls

Audit Trails

13. OBSERVABILITY

OpenTelemetry

Metrics

Logs

Traces

Business KPIs

14. DEVSECOPS

GitHub Actions

Terraform

Kubernetes

Policy-as-Code

Security Scanning

15. DEPLOYMENT

Vercel Frontend

Kubernetes Backend

Neo4j Cluster

PostgreSQL Cluster

16. UI SPECIFICATIONS

Executive Dashboard

Knowledge Graph Explorer

Traceability Explorer

Agent Studio

Learning Academy

17. PRODUCT ROADMAP

Quarter 1 Foundation

Quarter 2 MVP

Quarter 3 Pilot

Quarter 4 Scale

18. RESOURCE MODEL

Enterprise Architects

Product Managers

AI Engineers

Software Engineers

Platform Engineers

Data Engineers

Security Engineers

19. BUDGET MODEL

Platform Costs

Cloud Costs

AI Costs

Engineering Costs

Operations Costs

20. PRODUCTION READINESS CHECKLIST

Architecture Approved

Security Validated

Performance Tested

Governance Defined

AI Controls Implemented

Disaster Recovery Tested

21. TARGET STATE

An autonomous, AI-native enterprise platform capable of continuous evolution and measurable value realization.